

DECISION TREE ASSIGNMENT

<u>Machine</u>	<u>Temperature</u>	<u>Vibration</u>	<u>Failure</u>
1	High	High	Yes
2	High	Low	Yes
3	Medium	High	Yes
4	Low	Low	No
5	Low	Medium	No
6	Medium	Low	No
7	High	Medium	Yes
8	Low	High	No

Total Entropy

Variable: Failure

Yes = 4

No = 4

Total = 8

$$\begin{aligned}
 \text{Entropy} &= -P_1 \log_2 P_1 - P_2 \log_2 P_2 \\
 &= -4/8 \log_2 (4/8) - 4/8 \log_2 (4/8) \\
 &= -1/2 (-1) - 1/2 (-1) \\
 &= 1
 \end{aligned}$$

Information gain for temperature

① Temperature = Low

Yes = 0

No = 3

$$\text{Entropy} = -\log_2 3/5 = 0$$

② Temperature = Medium

Yes = 1

No = 1

$$\text{Entropy} = -1/2 \log_2 1/2 - 1/2 \log_2 1/2 = 1$$

③ Temperature = High

Yes = 3

No = 0

$$\text{Entropy} = -\log_2 3/3 = -\log_2 (1) = 0 //$$

Weighted Entropy for temperature

$$\begin{aligned}\text{Entropy (s, temperature)} &= 3/8(0) + 2/8(1) + 3/8(0) \\ &= 2/8 = \underline{\underline{0.25}}\end{aligned}$$

④ Information Gain for temperature

$$\text{I.G} = \text{Entropy (s)} - \text{weighted Entropy (Temp)}$$

$$= 1 - 0.25 = \underline{\underline{0.75}}$$

⑤ Information gain for vibration

① vibration = High

Yes = 2

No = 1

$$\text{Entropy} = -2/3 \log_2 (2/3) - 1/3 \log_2 1/3 = \underline{\underline{0.918}}$$

② Vibration = Medium

Yes = 1

No = 1

$$\text{Entropy} = -1/2 \log_2 1/2 - 1/2 \log_2 1/2$$

$$= 1 //$$

③ Vibration = Low

Yes = 1

No = 2

$$\text{Entropy} = \underline{\underline{0.918}}$$

⑥ Weighted Entropy (Vibration)

$$W \text{ Entropy} = \frac{3}{8}(0.918) + \frac{2}{8}(1) + \frac{3}{8}(0.918)$$

$$= 0.344 + 0.25 + 0.344$$

$$= \underline{\underline{0.938}}$$

$$\therefore \text{Info Gain} = \text{Entropy} - W \cdot \text{entropy}$$

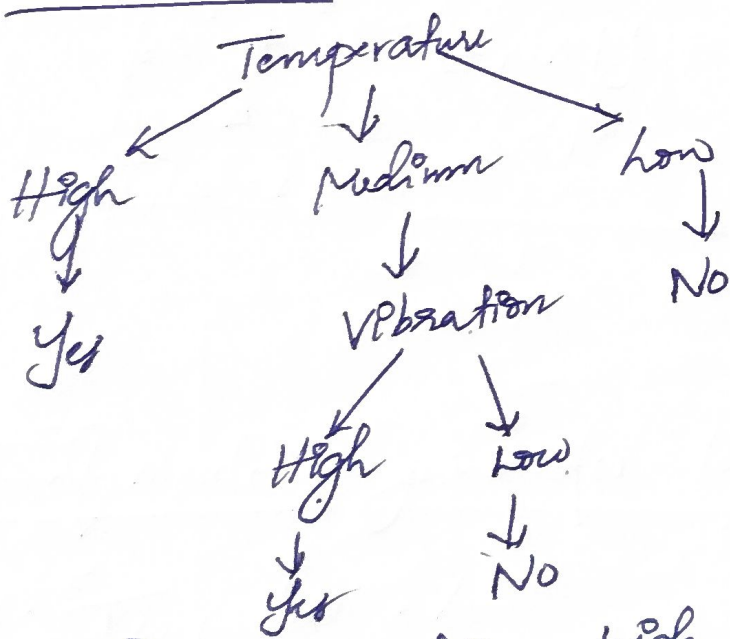
$$\underline{\text{Compare information gain}} = \underline{\underline{0.062}}$$

$$I.G(T) = 0.75$$

$$I.G(V) = 0.062$$

$\therefore$ Higher priority = Temperature

Decision Tree



$\therefore$ for temperature = high
vibration = medium

when temp is high $\rightarrow$ machine will fail

because

